Rishika Shreya Fadi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.13
Vedantha College	Class XII (PUE)	2024	94.00%
BGS NATIONAL PUBLIC SCHOOL	Class X (CBSE)	2022	97.00%

SCHOLASTIC ACHIEVEMENTS

- Achieved a peak rating of **1516 (Specialist)** on Codeforces, solved **400+** questions across different platforms. (Handle)
- Attained **Global Rank 389** and **India Rank 81** in **Codeforces Round 1106 (Div 2)** among 22,000+ participants.

PROJECTS

Course Recommendation Platform | *React, Python, FastAPI, PostgreSQL* | Github Live Demo

- Built a full stack course recommendation platform for **IIT Hyderabad students** using a **React** frontend, Go backend, **PostgreSQL** (Supabase), and a **Python FastAPI ML microservice**.
- Developed a **hybrid recommendation system** using rule based filtering, collaborative filtering, and a **Random Forest regressor** to **generate personalized elective recommendations** based on CGPA and course review metrics.
- Engineered REST APIs for authentication, course reviews, user activity, and recommendations using JWT authentication, bcrypt password hashing, and weighted SQL aggregation for course ratings.
- Designed a fault tolerant microservice architecture with fallback to database driven recommendations during ML service outages, and deployed the frontend, backend, and ML services independently on Vercel and Render.

RISC-V Pipeline & Cache Simulator | *C++, Bash* | Github

- Built a cycle-accurate **5-stage RISC-V pipeline simulator** supporting the RV64I, F, and D instruction set extensions, with modular execution modes for hazard detection, forwarding, branch prediction, and cache simulation.
- Reduced **control hazard penalties** by **half** through early branch resolution and static/dynamic 1-bit branch prediction.
- Implemented a **configurable cache simulator** supporting **LRU**, **FIFO**, and **Random** replacement policies for comparative memory & performance analysis.
- Developed an automated testing framework to validate simulator correctness and benchmark execution performance using CPI, throughput, latency, and branch prediction accuracy.

Gurujal NGO Platform | *React.js, Tailwind CSS* | Github

- Developed the frontend for **Gurujal**, an **NGO platform** promoting **Rainwater Harvesting (RWH)**, implementing responsive user interfaces across the calculator, vendor, and admin modules.
- Built reusable React components and multi-step workflows for the RWH calculator, vendor onboarding, authentication, and dashboard pages using **Tailwind CSS**.
- Integrated frontend with REST APIs to support the calculator, user authentication, location inputs, dynamic dashboards.
- Implemented responsive layout, client-side validation, and state management to deliver a consistent UX across devices.

KMap Minimizer | *React.js, TypeScript, Vite, Mermaid.js* | Github Live Demo

- Built an interactive web application for **Karnaugh Map simplification** supporting **2-5 variables**, featuring truth table editing, Gray-code visualization, and responsive React UI.
- Implemented the **Quine-McCluskey algorithm** to generate **minimized SOP expressions** through prime implicant generation and essential prime implicant selection, with POS generation via De Morgan's transformations.
- Developed logic gate visualization using **Mermaid.js** and generated Verilog modules from simplified expressions.
- Architected the frontend using React Context, custom hooks, and modular components, while eliminating prop drilling.

xv6 Operating System Extension | *C, xv6, OS Arch*

- Extended the **xv6 kernel** by implementing a **System Call Aware Multilevel Feedback Queue (SC-MLFQ)** scheduler with dynamic time quanta, syscall aware demotion, starvation prevention, and scheduler instrumentation.
- Built a demand paged **virtual memory manager** featuring lazy allocation, scheduler aware Clock page replacement, disk backed swap space, and per-process virtual memory statistics.
- Implemented a priority-aware disk subsystem supporting FCFS and SSTF scheduling policies together with Software RAID 0, RAID 1, and RAID 5 using XOR-based parity reconstruction.
- Developed comprehensive kernel test suites to validate scheduling fairness, page replacement, swap correctness, RAID integrity, disk scheduling, and system call behavior.

Arbitrary Precision Arithmetic Library | *Java, Maven, Docker* | Github

- Built a **Java library** and **CLI** tool for arbitrary precision (int/ float) arithmetic without relying on **BigInteger** or **BigDecimal**.
- Implemented **string based arithmetic algorithms** with sign magnitude representation, supporting addition, subtraction, multiplication, and division on arbitrarily large numbers.
- Designed floating point arithmetic by extending int operations with decimal scaling, normalization, and precision control.
- Packaged the library as a reusable **Maven JAR** and containerized it with **Docker** for portable cross platform execution.

SKILLS

- **Programming Languages** : C++, Python, SQL, Assembly (RISC-V).
- **Tools & Frameworks** : Git, Github, PostgreSQL, React.js, Vite, Tailwind CSS, Pandas, NumPy.

RELEVANT COURSES

- **Computer Science:** OS, Compilers, Data Structures, DBMS, Computer Architecture, Computation Theory, Algorithms, Software Development Fundamentals.
- **Mathematics**: Probability & Random Variables, Linear Algebra, Discrete Maths, Introduction To Statistics, Calculus.
- **Certifications** : Google Cloud Digital Leader Learning Path - Google Cloud Skills Boost

POSITIONS OF RESPONSIBILITY

- **Core Member** at **Epoch** (**AI/ML** club at IITH).
- **Public Relations and Marketing Outreach Coordinator** at **FCC** (Finance and Consulting Club at IITH).